Absent Classes, Not Client Drift: Diagnosing and Correcting Federated Collapse in Brain Tumor MRI Classification

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***Abstract*—Federated learning allows several hospitals to train a shared diagnostic model without exchanging patient images, but its accuracy degrades when the label distribution differs across sites. In brain tumor classification this degradation is severe, because a participating centre may never encounter one of the tumor types at all. We show that under a Dirichlet partition with concentration 0.3, thirteen of seventeen simulated sites hold fewer than three tumor types, and that a federated model trained with the standard cross-entropy loss under these conditions loses six to seven points of balanced accuracy and becomes acutely sensitive to how the partition happens to fall, its worst partitions returning 0.447 against a chance level of 0.333. We trace the failure to a specific term: for a class a client does not hold, the cross-entropy gradient on that class logit is strictly positive at every step, so local training displaces the corresponding output parameters in a direction fixed by the partition draw. Removing that term entirely, by masking absent-class logits during local updates, raises balanced accuracy from 0.584 to 0.649 over twenty seeds (Holm-corrected $p = 0.002$, Cohen's $d = 0.91$), reduces the across-seed standard deviation from 0.074 to 0.028, and closes 54% of the gap to centralised training on the pooled data. The advantage replicates on the BRISC 2025 benchmark, which uses independently annotated masks and a different class balance. Comparison against logit calibration shows the two corrections to be statistically indistinguishable ($p = 0.12$), which supports the diagnosis rather than any single remedy: what matters is that the absent-class term is removed, not how. A controlled sweep confirms this, with the benefit scaling from 17.5 points at high local absence to no measurable effect when every client holds every class.**

***Index Terms*—Federated learning, label distribution skew, restricted softmax, brain tumor, magnetic resonance imaging, non-IID data, SCAFFOLD.**

I. Introduction

MAGNETIC resonance imaging is the primary modality for characterising intracranial tumors, and automated classification of tumor type from MRI has been studied extensively as a decision-support aid [1]–[3]. Progress in this area depends on access to large annotated cohorts, yet the images involved are among the most tightly regulated categories of clinical data. Institutional review requirements and national privacy legislation frequently prevent hospitals from pooling scans into a central repository, which leaves individual centres training on datasets too small to support a reliable model.

Federated learning offers a way around this constraint. Rather than moving images, participating sites exchange model updates with a coordinating server, so that patient data never leaves the institution where it was acquired [4], [5]. The approach has been evaluated for multi-institutional brain tumor segmentation [6] and has been proposed as a general framework for collaborative clinical machine learning [7].

The difficulty is that the classical convergence analysis of federated averaging assumes that client datasets are drawn from a common distribution, and clinical data violate this assumption in a specific and consequential way. Referral patterns, departmental specialisation and scanner availability determine which conditions a centre sees. A neurosurgical unit that treats pituitary adenomas may accumulate hundreds of such cases and none of another tumor type. The resulting label skew is not a mild statistical nuisance: when a client holds no examples of class c, the gradient of the cross-entropy loss with respect to the logit of c is strictly positive at every optimisation step, so local training drives that logit downward without any counteracting signal. Averaging models that have each suppressed a different subset of classes yields a global classifier whose final layer has been corrupted in a manner that depends on how the partition happened to fall.

Several corrections operate directly on the softmax. FedRS rescales the logits of absent classes by a factor $\alpha \in [0, 1]$ during local training [8]; FedLC generalises this to a per-class margin derived from label frequency [9]; FedVLS adds a distillation term aimed specifically at vacant classes [10]. These methods were developed and tuned on natural-image benchmarks with large label spaces, where each client is missing most of the classes. The regime encountered in tumor-type classification is different: the label space contains three classes, and a client typically misses one. Whether the tuned intermediate values of α reported for CIFAR-scale problems remain appropriate when the label space is small has not been examined, and the behaviour of the limiting case $\alpha = 0$ has not been characterised.

This paper makes the following contributions.

- We identify the mechanism behind federated collapse in this setting. The cross-entropy gradient on a class the client does not hold is strictly positive at every step, so local training displaces the corresponding output parameters along a direction determined by the partition draw. We show that this term, and not client drift in the shared representation, is what destabilises training, and we quantify the condition that produces it: a Dirichlet partition with concentration 0.3 leaves 13 of 17 clients without at least one tumor type.

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- We show that removing the term is what matters, not the particular way it is removed. Full masking of absent-class logits and frequency-based logit calibration both recover 6 to 7 points of balanced accuracy over federated averaging and are statistically indistinguishable from each other ($p = 0.12$ over twenty seeds), while partial scaling at the setting published for FedRS recovers less than one point. This locates the effect in the presence or absence of the term rather than in any specific operator, and the comparison against calibration is not reported in either originating paper.
- We place the result on an absolute scale by measuring centralised training on the pooled data. The correction closes 54% of the gap between federated averaging and the centralised upper bound, which no prior study of this dataset reports.
- We isolate the mechanism through a controlled sweep over partition severity. The benefit scales monotonically with the mean fraction of classes absent per client, from 17.5 points at high absence to no measurable effect when every client holds every class, which is the behaviour the gradient argument predicts and a general regularisation account does not.
- We report three negative findings that follow from the same analysis: learning-rate warmup with cosine decay and extended training, both standard remedies for unstable federated optimisation, increase the collapse rate under this form of skew; the FedProx proximal term adds nothing to the corrected objective; and the benefit halves when the backbone is enlarged from a three-block convolutional network to a residual network, which bounds the regime in which the correction is worth applying.

II. Related Work

A. Federated Optimisation under Heterogeneous Data

Federated averaging aggregates locally trained weights in proportion to dataset size [4]. Its accuracy deteriorates when client distributions diverge, an effect analysed empirically by Zhao *et al.* [11] and confirmed across a range of partitions by Li *et al.* [12]. Two families of correction dominate. FedProx constrains the local solution to remain near the current global model through a proximal penalty [13]. SCAFFOLD instead estimates the difference between local and global gradient directions and subtracts it during local steps, which removes client drift without restricting the size of the local update [14]. Both target drift in the shared representation. Neither addresses the case in which a client's local objective is defined over a strict subset of the label space, which is the situation examined here.

Normalisation interacts with heterogeneity. Batch normalisation estimates statistics from local batches, so under non-identical client distributions the running estimates diverge and averaging becomes unreliable. Hsieh *et al.* showed that replacing batch normalisation with group normalisation [15] recovers most of the accuracy lost to this effect [16], a result reproduced in later normalisation benchmarks [17]. FedBN takes the alternative route of keeping normalisation layers local [18]. We adopt group normalisation for the reasons given in [16].

B. Softmax and Logit Adjustment for Label Skew

FedRS is the closest antecedent to the present work [8]. It multiplies the logits of classes absent from a client by a scalar $\alpha \in [0, 1]$, which reduces but does not eliminate the gradient reaching those logits; $\alpha = 1$ recovers federated averaging. The reported experiments use CIFAR-10 and CIFAR-100 with $\alpha = 0.5$, and gains over federated averaging range from 4.6 to 14.9 points depending on the partition and the backbone. FedLC replaces the single scalar with a per-class margin computed from local label frequencies, treating missing classes as the extreme of the minority-class case, and reports larger gains on Dirichlet partitions of CIFAR-10 [9]. FedVLS observes that calibration methods can fall below federated averaging when the number of vacant classes is large, and adds a distillation objective to compensate [10].

C. Relation to the Boundary Case Already Reported in FedRS

Because our correction is the boundary case of FedRS, we set out explicitly what that paper already covers, so that the difference is a matter of record rather than of emphasis.

Li and Zhan evaluate $\alpha \in \{0.0, 0.1, 0.5, 0.9, 1.0\}$, so the value we use is among those they tried. They also treat it analytically, observing that at $\alpha = 0$ the method "degenerates into fixed proxies" whose weights are not updated and therefore do not contribute to the aggregation, which is the same mechanism we describe in Section III-B. Their empirical summary is that $\alpha = 0$ obtains results comparable to $\alpha = 0.5$ in most cases and can be worse with more local epochs. Consistent with that reading, $\alpha = 0$ appears in their convergence figures but not in their headline tables, and the setting they carry forward is $\alpha = 0.5$. They further report a combined variant, Scaffold/RS, which is the best configuration in their comparison, so pairing the restricted objective with SCAFFOLD is likewise established rather than new. Finally, in an appendix they observe weaker improvements with a VGG11 backbone than with a smaller network and conjecture that the restriction helps more when the feature extractor is small; the backbone measurement in Section V-F is the controlled version of that conjecture.

Three things are therefore not claimed here: the operator, the pairing with SCAFFOLD, and the qualitative idea that backbone size matters.

What differs is the regime and, with it, the answer. Their evaluation uses ten to one hundred classes with two to twenty per client, where a client is missing most of the label space and removing all of that negative evidence costs as much as the protection gains; "comparable to $\alpha = 0.5$" is the expected outcome there. Our setting has three classes with a mean local absence of 0.31, so a client typically lacks one class, the discarded evidence is small, and the protection dominates. In that regime $\alpha = 0$ is not comparable to $\alpha = 0.5$ but ahead of it by 5.8 points (Table II). Their own caveat is consistent with

this: the case they found unfavourable is many local epochs, and we use one.

The contribution is accordingly not the boundary value but an account of what governs it. Section V-E measures the benefit as a function of local class absence and finds it monotone, ranging from 17.5 points to nothing, which turns a reported equivalence and a measured advantage into two points on one curve rather than a contradiction. To that we add a comparison against logit calibration, which neither paper reports and which shows the two removals to be indistinguishable; an absolute reference from centralised training; and patient-level evaluation on medical images, where the missing-class condition arises from referral patterns rather than from a synthetic partition.

D. Brain Tumor Classification and Evaluation Practice

The Figshare dataset introduced by Cheng *et al.* [1] is the standard benchmark for three-class tumor-type classification and remains the reference point for reported accuracy. Subsequent work has applied engineered feature descriptors [2], transfer learning from natural-image networks [3], [19] and capsule architectures [20].

Comparison across these studies is complicated by evaluation protocol. Each patient contributes multiple adjacent slices with near-identical appearance, so a random slice-level split places highly correlated images in both training and test sets. Yagis *et al.* measured the consequence directly on brain MRI cohorts for Alzheimer's and Parkinson's disease, and found that slice-level splitting inflated reported accuracy by 29 to 55 points relative to subject-level splitting on the same data, and that randomly labelled scans reached approximately 96% under a slice-level split against 50% under a subject-level split [21]. The pathology differs from ours, but the mechanism is a property of the data structure rather than of the disease: any cohort in which one subject contributes many adjacent slices is exposed to it. We therefore group by patient identifier throughout and treat published slice-level figures as measuring a different quantity.

III. Methods

A. Problem Setting

Let K clients hold private datasets $\mathcal{D}_1, \dots, \mathcal{D}_K$ over a common label space $\mathcal{Y}$ with $|\mathcal{Y}| = C$. Write $\mathcal{P}_k \subseteq \mathcal{Y}$ for the set of classes that appear in $\mathcal{D}_k$ and $\mathcal{A}_k = \mathcal{Y} \setminus \mathcal{P}_k$ for the absent classes. Federated training minimises

$$\min_{\theta} \sum_{k=1}^{K} \frac{n_k}{n} F_k(\theta), \qquad F_k(\theta) = \frac{1}{n_k} \sum_{(x,y)\in\mathcal{D}_k} \ell(f_\theta(x), y), \tag{1}$$

with $n_k = |\mathcal{D}_k|$ and $n = \sum_k n_k$. We define the *local absence* of client k as $a_k = |\mathcal{A}_k|/C$ and report the mean $\bar{a} = K^{-1} \sum_k a_k$ as the summary statistic of partition severity.

B. The Absent-Class Gradient

Let $z = f_\theta(x) \in \mathbb{R}^C$ denote the logits and $p = \mathrm{softmax}(z)$. For the cross-entropy loss $\ell(z, y) = -\log p_y$, the derivative with respect to the logit of an arbitrary class c is

$$\frac{\partial \ell}{\partial z_c} = p_c - \mathbb{1}[c = y]. \tag{2}$$

If $c \in \mathcal{A}_k$ then $\mathbb{1}[c = y] = 0$ for every sample in $\mathcal{D}_k$, so the derivative reduces to $p_c > 0$ at every step. The parameters of the output unit for class c are therefore driven in one direction throughout local training, and the magnitude of this displacement depends on how many and which classes are absent, which is determined by the partition and thus by the random seed. When the server averages models whose output layers have been displaced along different class directions, the resulting classifier is biased in a way that varies from run to run. This accounts for both the loss of mean accuracy and the large across-seed variance reported in Section V.

C. Class-Presence-Guided Restricted Softmax

The correction rescales the contribution of absent classes to the local partition function. For client k,

$$\ell_k(z, y) = -\log \frac{\exp(z_y)}{\sum_{c\in\mathcal{Y}} s_c^{(k)} \exp(z_c)}, \qquad s_c^{(k)} = \begin{cases} 1, & c \in \mathcal{P}_k, \\ \alpha, & c \in \mathcal{A}_k, \end{cases} \tag{3}$$

which recovers standard cross-entropy at $\alpha = 1$ and reproduces FedRS [8] for intermediate α. Differentiating (3) with respect to an absent-class logit gives

$$\frac{\partial \ell_k}{\partial z_c} = \frac{\alpha \exp(z_c)}{\sum_{c'} s_{c'}^{(k)} \exp(z_{c'})} = \alpha\, \tilde{p}_c, \qquad c \in \mathcal{A}_k, \tag{4}$$

so the scaling factor acts directly on the rate at which the absent-class parameters are displaced. Setting $\alpha = 0$ removes the term entirely: the absent-class output parameters receive no gradient and are returned to the server unchanged from the values at which they were broadcast. In our implementation this is realised by adding $-\infty$ (numerically -10^9) to the absent logits before the softmax, which is equivalent to (3) in the limit and numerically stable.

Intermediate values of α slow the displacement without stopping it. In a federated setting the effect accumulates, because each round re-broadcasts the aggregated parameters and repeats the local pass. Over T rounds the residual displacement scales with αT, so a factor that appears small in a single epoch is not small after forty rounds. This is the reason we expect, and observe, that $\alpha = 0$ dominates intermediate settings when local absence is high.

The argument also predicts when the correction should make no difference. If $\mathcal{A}_k = \emptyset$ for all k, the modified loss (3) is identical to cross-entropy for any α, and no effect of the correction should be measurable. Section V-E tests this prediction directly.

Algorithm 1 Class-Presence-Guided Federated Training

Input: clients $1..K$, rounds T, learning rate η, scaling α
Server initialises θ^0 and control variate c^0
for $t = 0$ **to** $T - 1$ **do**
  Server broadcasts θ^t, c^t
  for each client k **in parallel do**
    $\mathcal{P}_k \leftarrow$ set of classes present in $\mathcal{D}_k$
    $\theta_k \leftarrow \theta^t$
    for each local batch (x, y) **do**
      $g \leftarrow \nabla_{\theta_k} \ell_k(f_{\theta_k}(x), y)$ using (3)
      $g \leftarrow \mathrm{clip}\big(g + c^t - c_k,\ 1\big)$
      $\theta_k \leftarrow \theta_k - \eta g$
    end for
    Update c_k; return $\Delta_k = \theta_k - \theta^t$
  end for
  $\theta^{t+1} \leftarrow \theta^t + \sum_k (n_k/n)\, \Delta_k$
  $c^{t+1} \leftarrow c^t + K^{-1} \sum_k (c_k - c^t)$
end for

D. Training Procedure

Local optimisation uses SCAFFOLD [14], whose control variates correct drift in the shared feature extractor, combined with the restricted objective (3), which corrects the output layer. The two act on different parts of the model and are complementary. All normalisation layers are group normalisation [15] following [16]. Local sampling is class-balanced within each client, and gradients are clipped to unit norm. Algorithm 1 summarises the procedure.

E. Evaluation Protocol

Test patients are held out before partitioning, and all slices belonging to a patient are assigned to the same side of the split, so no patient contributes to both training and evaluation [21].

The primary metric is balanced accuracy, the unweighted mean of per-class recall. The class distribution of the Figshare dataset is uneven (708 meningioma, 1426 glioma, 930 pituitary), so a classifier that predicts the majority class for every input attains 46.5% plain accuracy while contributing nothing diagnostically; the same classifier scores 0.333 under balanced accuracy. Since the failure mode under study is precisely collapse onto a single class, plain accuracy would obscure the phenomenon the experiments are designed to measure. We define collapse as a run with balanced accuracy at or below 0.40.

Every configuration is run over 20 seeds, with the seed controlling both the Dirichlet draw and model initialisation, so that comparisons are paired. Differences are assessed with the Wilcoxon signed-rank test, which does not assume normality of the per-seed differences, and reported with Cohen's d for paired samples and a 10,000-resample bootstrap confidence interval. Because each method is compared against three baselines, p-values are adjusted with the Holm step-down procedure [23].

IV. Experimental Setup

A. Datasets

The primary dataset is the Figshare collection of Cheng *et al.* [1], comprising 3064 contrast-enhanced T1-weighted slices from 233 patients, distributed as 708 meningioma, 1426 glioma and 930 pituitary, with expert tumor masks and patient identifiers.

External validation uses BRISC [22], released with expert segmentation masks for the same three tumor types. We use 3000 slices, 1000 per class, drawn from the segmentation subset. BRISC does not distribute patient identifiers, so clients are formed by drawing slices rather than patients; this limitation is stated again in Section VII. We also note that BRISC is assembled in part from sources that include the Cheng collection, so the two benchmarks are not fully independent. The masks are newly annotated and the class balance differs, which makes the replication informative as a robustness check rather than as evidence of external generalisation in the strict sense.

B. Partitioning and Model

Training patients are distributed over $K = 18$ clients by drawing, for each class, a vector of proportions from a Dirichlet distribution with concentration α_{dir}. Clients receiving fewer than four samples are removed, which leaves 17 active clients at $\alpha_{\mathrm{dir}} = 0.3$. At this setting 13 of 17 clients hold fewer than three tumor types, and $\bar{a} = 0.31$. Fig. 1 shows the resulting distribution of absent classes and the composition of each client.

The classifier is a three-block convolutional network with 16, 32 and 64 channels, group normalisation after each convolution, global average pooling and a linear output layer. Images are resized to 128×128 and scaled to $[0, 1]$. Local optimisation uses stochastic gradient descent with learning rate 0.01, one local epoch per round and a batch size of 32; training runs for 40 rounds. The deliberately small backbone keeps the comparison focused on the loss function rather than on representational capacity. Section V-F measures what changes when the backbone is enlarged.

Two notation conventions are used throughout: α_{dir} denotes the Dirichlet concentration governing partition severity, and α without subscript denotes the restricted-softmax scaling factor of (3).

C. Reproducibility

Every configuration is executed from a fixed seed that determines the Dirichlet draw, the train and test assignment of patients, the model initialisation and the order of local batches. Methods compared within a table therefore see identical partitions, which is what makes the paired tests in Section III-E valid. Results are written incrementally to disk so that interrupted runs resume without repeating completed cells, and the aggregate tables are regenerated from those records rather than transcribed. Statistical tests are computed from the per-seed values, not from the summary statistics.

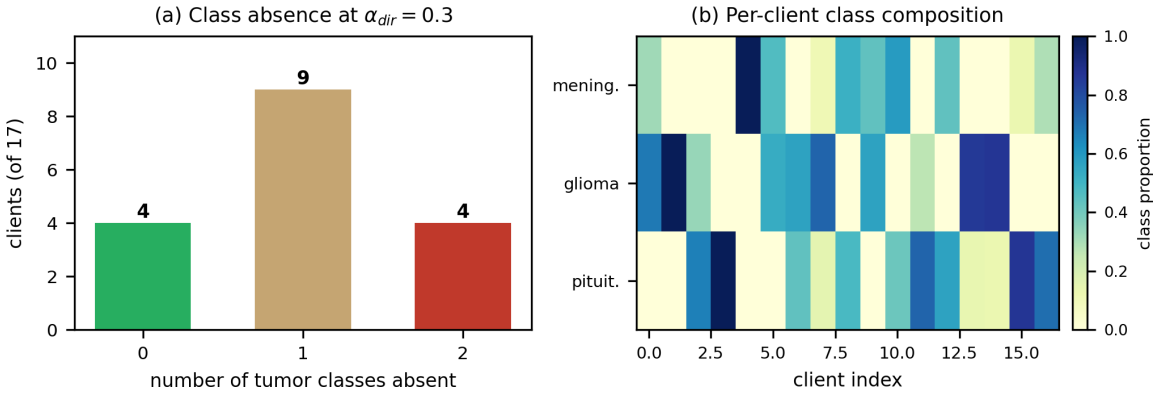


Fig. 1. The missing-class condition at $\alpha_{\mathrm{dir}} = 0.3$. (a) Only four of seventeen clients hold all three tumor types; nine are missing one and four are missing two. (b) Class composition of each client, where empty cells indicate a tumor type the client never observes during training.

We note one methodological check that shaped the design of the evaluation. An earlier version of this study assessed per-patient adaptation using a support and query split of each held-out patient's slices. Because a patient carries a single tumor type across all slices, the support set determines the label of the query set, and a predictor that ignores the images entirely and copies the majority support label attains a balanced accuracy of 1.000. Per-patient adaptation is therefore not measurable on this task, and we report no such figures.

V. Results

A. Which Interventions Stabilise Training

We first isolate the effective component by enabling one intervention at a time on top of an uncorrected federated baseline (Table I and Fig. 2). This ablation was run at an earlier and shorter schedule than the comparisons that follow, twenty rounds with two local epochs rather than forty rounds with one, so its baseline sits at 0.551 rather than the 0.584 reported in Table II. The two numbers are not in conflict; they describe the same objective under different budgets, and only within-table comparisons are made. Gradient clipping alone does not help. Learning-rate warmup with cosine decay and extended training, both standard responses to unstable federated optimisation, increase the collapse rate from zero to 15 and 14 runs out of 20 respectively. Class-balanced local sampling leaves the mean essentially unchanged. The restricted softmax raises balanced accuracy from 0.551 to 0.611 and eliminates collapse. Adding the FedProx proximal term on top of the corrected objective changes neither the mean nor the variance.

The two configurations that fail share a property: both increase the number of local optimisation steps taken against the uncorrected objective, either by lengthening the schedule or by raising the effective step size in the later rounds. Under (2), more steps against that objective means more displacement of the absent-class parameters, so the direction of these results follows from the mechanism rather than contradicting it.

TABLE I
Component ablation on FIGSHARE over 20 seeds. Collapse denotes a run with balanced accuracy ≤ 0.40.

Configuration	Balanced acc.	Collapses
Federated baseline	0.551 ± 0.031	0/20
+ gradient clipping	0.529 ± 0.052	1/20
+ warmup, cosine decay	0.371 ± 0.065	15/20
+ extended training	0.379 ± 0.068	14/20
+ balanced sampling	0.544 ± 0.081	1/20
+ restricted softmax	$\mathbf{0.611 \pm 0.055}$	0/20
+ FedProx term	0.611 ± 0.055	0/20

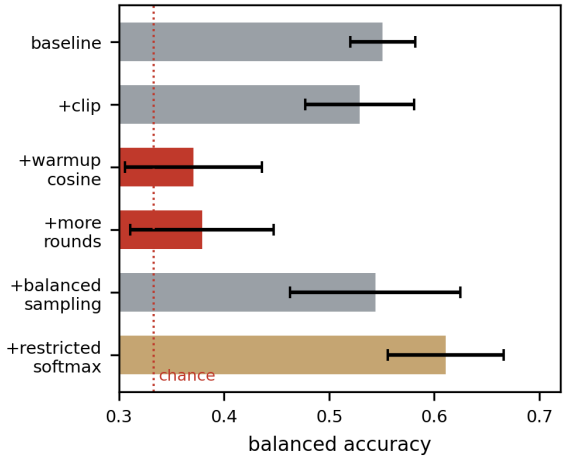


Fig. 2. Component ablation. Only the restricted softmax (bottom) improves on the uncorrected baseline. Learning-rate warmup with cosine decay and extended training push the mean toward the chance level of 0.333 (dotted line).

B. Comparison against Softmax Corrections

Table II compares four local objectives on identical partitions. Federated averaging and FedRS at its published setting $\alpha = 0.5$ behave almost alike, separated by less than one point. The two objectives that remove the absent-class term outright, full masking and frequency-based logit calibration, both gain six to seven points over federated averaging, and the difference between them is not significant ($\Delta = 0.004$, Holm $p = 0.108$,

TABLE II
FIGSHARE, PATIENT-LEVEL BALANCED ACCURACY OVER 20 SEEDS. DIFFERENCES ARE COMPUTED AGAINST THE HARD MASK; p-VALUES ARE HOLM-ADJUSTED ACROSS THE THREE COMPARISONS. THE CENTRALISED BOUND IS TRAINED ON THE POOLED DATA WITH THE SAME BACKBONE AND STEP BUDGET (5 SEEDS).

Method	Balanced acc.	Δ	p	d
FedAvg	0.584 ± 0.074	-0.065	0.002	0.91
FedRS, $\alpha = 0.5$	0.592 ± 0.072	-0.058	0.002	0.86
FedLC	0.653 ± 0.026	$+0.004$	0.123	0.34
Hard mask, $\alpha = 0$	$\mathbf{0.649 \pm 0.028}$	—	—	—
Centralised bound	0.705 ± 0.011	—	—	—

masking ahead on 8 of the 20 partitions and calibration on the other 12).

We read this pattern as evidence for the diagnosis rather than for a particular operator. Partial scaling attenuates the absent-class gradient by a factor α but leaves it in the objective, and over forty rounds the residual displacement accumulates; the result is close to no correction at all. Once the term is removed, whether by masking or by a margin large enough to be equivalent to masking, the method used to remove it makes no measurable difference. The practical distinction between the two is that masking requires no hyperparameter, whereas calibration introduces a temperature that must be tuned; in a federated clinical deployment, where no held-out validation set spans all sites, this is a meaningful simplification.

The reduction in across-seed standard deviation is as informative as the change in the mean. Federated averaging and FedRS vary with standard deviation 0.074 and 0.072, while the two corrected objectives sit at 0.028 and 0.026. The worst partition under the hard mask returns 0.598, above the *mean* of either uncorrected baseline.

Trained on the pooled data, the same backbone reaches 0.705. The correction therefore closes 54% of the distance between federated averaging and centralised training, which locates the remaining gap in the federation itself rather than in the loss function.

C. Distribution of Outcomes across Partitions

The aggregate figures in Table II conceal a difference in the shape of the distributions. Federated averaging and FedRS produce a bimodal pattern: on most partitions they reach the mid-0.6 range, but on a minority they fall to the high 0.4s. Seeds 5, 13 and 14 illustrate this, returning 0.449, 0.488 and 0.447 under federated averaging and 0.494, 0.487 and 0.447 under FedRS. On the same three partitions the hard mask returns 0.689, 0.636 and 0.670, and calibration returns 0.682, 0.632 and 0.661. The two corrected objectives track each other closely on precisely the partitions where the uncorrected ones fail, which is the pattern the diagnosis predicts. The lowest result over all twenty seeds under the hard mask is 0.598, above the mean of either uncorrected baseline. Fig. 3 shows the paired outcomes seed by seed, and Fig. 4 summarises the resulting spread on both benchmarks.

This pattern is what the mechanism predicts. The seeds on which the baselines fail are those whose Dirichlet draw concentrates entire classes on a small number of clients, so that the remaining clients carry a large absent-class gradient. Since the hard mask removes that gradient regardless of how the draw falls, its outcome depends much less on the partition. Reporting only the mean would understate the difference between the methods, because the methods differ most in the tail.

The hard mask wins on 15 of 20 paired comparisons against FedRS on Figshare. The five partitions on which it does not win are all cases where every method performs well, and the differences there are within 0.02. No partition was found on which the hard mask fails while a baseline succeeds.

TABLE III
BRISC, BALANCED ACCURACY OVER 20 SEEDS. DIFFERENCES ARE COMPUTED AGAINST THE HARD MASK, HOLM-ADJUSTED.

Method	Balanced acc.	Δ	p	d
FedAvg	0.621 ± 0.061	-0.055	0.0015	0.87
FedRS, $\alpha = 0.5$	0.624 ± 0.062	-0.052	0.0015	0.82
FedRS, $\alpha = 0.25$	0.635 ± 0.061	-0.041	0.0018	0.77
Hard mask, $\alpha = 0$	$\mathbf{0.676 \pm 0.046}$	—	—	—

TABLE IV
BALANCED ACCURACY AS A FUNCTION OF PARTITION SEVERITY AND SCALING FACTOR, EIGHT SEEDS PER CELL. THE FINAL COLUMN REPORTS THE MARGIN OF $\alpha = 0$ OVER $\alpha = 0.5$ WITH ITS PAIRED WILCOXON p-VALUE.

α_{dir}	$\bar{a}$	α=0	α=0.5	α=1	Margin	p
0.1	0.50	0.629	0.454	0.471	+0.175	0.008
0.5	0.31	0.670	0.607	0.602	+0.063	0.023
3.0	0.07	0.659	0.657	0.656	+0.002	0.836

D. Replication on BRISC

Table III repeats the comparison on BRISC. The ordering of the four methods is preserved and the margin over FedRS at $\alpha = 0.5$ is 5.2 points, close to the 5.8 observed on Figshare. No method collapses on this benchmark, which is consistent with its balanced class distribution, and the hard mask nevertheless retains its advantage. The correction therefore does more than prevent degenerate solutions.

E. The Margin Scales with Local Absence

Section III argued that the benefit of $\alpha = 0$ should depend on how many classes are absent from each client and should disappear when none are. We test this by varying α_{dir} over $\{0.1, 0.5, 3.0\}$, which produces mean local absence $\bar{a} \in \{0.50, 0.31, 0.07\}$, and evaluating $\alpha \in \{0, 0.5, 1\}$ at each level over eight seeds (Table IV).

The margin decreases monotonically with local absence and is no longer distinguishable from zero when clients hold nearly the full label space. We emphasise that $\alpha = 0$ remains the best of the three settings at every level tested; what changes is the size of the advantage, not its sign. The correction is therefore inactive rather than harmful in the balanced regime, which is the behaviour (3) predicts when $\mathcal{A}_k = \emptyset$.

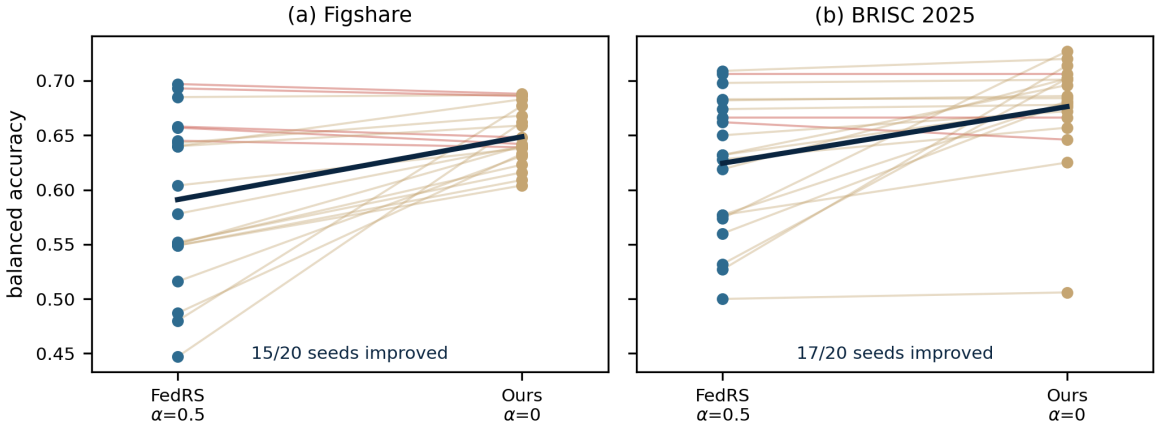


Fig. 3. Paired per-seed comparison against FedRS at $\alpha = 0.5$. Each line connects the two objectives on one partition; the heavy line joins the means. The partitions on which the uncorrected baseline falls below 0.55 are those with the highest local absence, and these are the partitions on which removing the absent-class term has the largest effect.

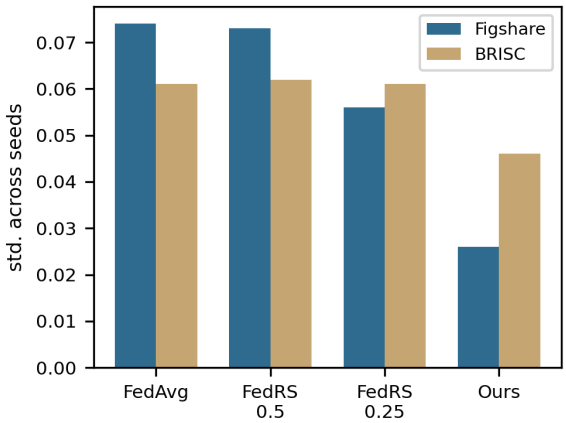


Fig. 4. Standard deviation of balanced accuracy across seeds. Objectives that remove the absent-class term give the narrowest spread on both benchmarks.

TABLE V
EFFECT OF BACKBONE CAPACITY, 10 SEEDS. GAINS ARE MEASURED AGAINST FEDERATED AVERAGING ON THE SAME BACKBONE.

	Three-block CNN		Residual network	
Method	Bal. acc.	Gain	Bal. acc.	Gain
FedAvg	0.584	—	0.559	—
FedRS 0.5	0.592	+0.8	0.563	+0.3
FedLC	0.653	+6.9	0.593	+3.4
Hard mask	0.649	+6.5	0.588	+2.9

F. Sensitivity to Backbone Capacity

The corrections studied here act on the output layer, so their effect should depend on how much of the classification is carried by that layer. We repeated the comparison with a residual network of roughly ten times the parameter count, keeping the partitions, budget and normalisation unchanged (Table V).

Two observations follow. The benefit of removing the absent-class term halves when the backbone is enlarged, from 6.5 to 2.9 points for the hard mask and from 6.9 to 3.4 for calibration. This is the behaviour expected of a correction that acts on the output layer: as more of the classification is carried by the feature extractor, the share attributable to the final layer falls, and with it the effect of altering that layer's objective. The measurement bounds the regime in which our result should be expected to hold.

The residual network is also worse in absolute terms on every objective. With roughly 2400 training slices divided among seventeen clients, the larger model has too little data per client to justify its capacity, and the ordering of the methods is preserved but compressed. We therefore report the three-block network as the primary configuration, a choice the comparison supports rather than assumes.

VI. DISCUSSION

The experiments support a claim about a mechanism rather than about a method. Federated training fails here because clients optimise an objective defined over classes they do not hold, and the size of the failure tracks how much of the label space is missing locally. Three results converge on this reading.

Partial and full removal of the absent-class term behave very differently. Scaling the term by $\alpha = 0.5$ recovers less than one point, while removing it recovers six to seven. If the benefit came from regularisation in a general sense, a factor of two in the coefficient would not produce a factor of seven in the outcome. What separates the two is that attenuation leaves a residual gradient which accumulates over rounds, whereas removal does not.

Two unrelated operators that both remove the term perform the same. Masking sets the absent-class contribution to zero directly; calibration subtracts a margin large enough to have the same effect. They differ in form, in the parameters they require, and in the literature they come from, yet their outcomes cannot be distinguished at twenty seeds. This is what one expects if the operative variable is the presence of the term rather than the identity of the correction, and it is why we frame the contribution as a diagnosis. A practitioner may

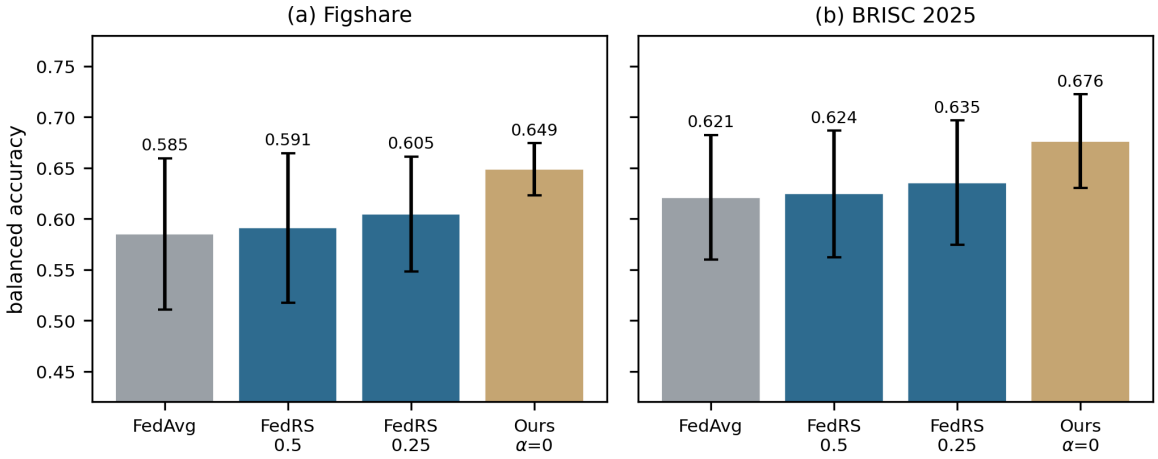


Fig. 5. Balanced accuracy over 20 seeds on both benchmarks. Error bars show one standard deviation across seeds. The ordering of methods is preserved between datasets, and the hard mask combines the highest mean with the smallest spread in each case.

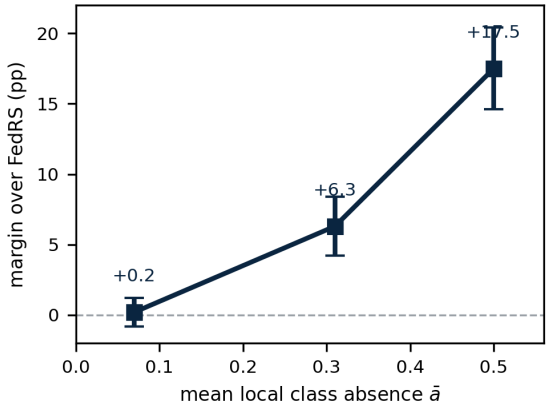


Fig. 6. Margin of $\alpha = 0$ over $\alpha = 0.5$ as a function of mean local class absence. Error bars show the standard error of the paired difference across eight seeds. The effect disappears when clients hold nearly the complete label space.

reasonably choose either; masking is simply the one that needs no tuning.

The benefit disappears exactly when the mechanism says it should. At $\bar{a} = 0.07$, where almost every client holds every class, the corrected and uncorrected objectives are within 0.2 points and the difference is not significant. A method that helped through some general smoothing effect would not switch off so cleanly.

The variance reduction deserves separate comment. The displacement of the absent-class parameters depends on which classes each client happens to lack, which is a property of the partition draw. Removing the term removes a seed-dependent error source, and the standard deviation falls from 0.074 to 0.028 while the mean rises by six points. For deployment this is arguably the more relevant quantity: a federation is assembled once and its participants are not resampled, so what matters is the accuracy obtained on the particular partition at hand, and the correction narrows the range of outcomes that partition can produce.

The negative results are consistent with the same account. Warmup with cosine decay and extended training both increase the number of local steps taken against an objective that still contains the absent-class term, which (2) identifies as the harmful quantity. That these are standard remedies for federated instability indicates that the failure mode here is distinct from the client drift they were designed for. FedProx points the same way: a proximal penalty limits how far the local solution travels but not the direction in which it travels, so it cannot prevent a displacement that is systematic rather than excessive.

VII. Limitations

Several constraints bound these conclusions.

The correction we recommend is not novel as an operator, and neither is its pairing with SCAFFOLD. Both appear in the paper that introduced the scaling rule, which evaluates the boundary value among its settings and reports a combined Scaffold/RS variant; logit calibration was proposed separately as a generalisation of the same idea. Our claim concerns which property of these objectives does the work, and the regime in which the boundary case is the right choice rather than merely an equivalent one. Readers seeking a new operator will not find one here. Section II-C states the overlap in full.

Both datasets consist of contrast-enhanced T1-weighted slices, so the results do not speak to multi-sequence or volumetric protocols. The two benchmarks are also not fully independent: BRISC is assembled from sources that include the Cheng collection. Its masks are independently annotated and its class balance differs, which makes the replication a robustness check rather than validation on an unseen population.

BRISC does not distribute patient identifiers, so clients on that benchmark are formed by drawing slices. Slices from one patient may be split across clients and across the train and test partitions, which is the leakage condition avoided on Figshare. The BRISC figures are therefore comparable across methods, since every method faces the same partition, but not comparable in absolute terms to the patient-level Figshare figures.

The benefit halves with a larger backbone, as Table V shows. Our results are established for models of the size appropriate to the data available per client in this setting; they should not be extrapolated to federations with enough data per site to support a large feature extractor.

Balanced accuracy weights the three tumor types equally, which is appropriate for measuring discriminative capacity but does not reflect the asymmetric cost of diagnostic errors in clinical use. A cost-sensitive evaluation would require a clinically validated cost matrix, which we did not have access to.

Finally, we tested one alternative account and rejected it. An auxiliary contrastive term with targets derived from continuous fractal descriptors, intended to organise the latent space so that absent classes retain structure, gave a marginal and unstable gain when added to the mask and performed ten points worse when used in its place, with collapse on two partitions. The failure of a representation-space remedy is further evidence that the pathology resides in the output layer.

VIII. Conclusion

Federated brain tumor classification collapses under severe label skew because clients optimise a loss defined over classes they do not hold, which displaces the corresponding output parameters in a direction fixed by the partition draw. Removing that term raises balanced accuracy by six to seven points over federated averaging, cuts across-seed variance by more than half, and closes 54% of the gap to centralised training. Two different operators that remove it perform equivalently, while attenuating it recovers almost nothing, which places the effect in the term itself rather than in any particular remedy. The benefit scales with the fraction of the label space absent at each client and vanishes when no classes are missing. For federations of the size encountered in tumor-type classification, masking absent-class logits achieves this without introducing a hyperparameter.

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